PR-3

PART - A

① What is Logistic Regression and why is it suitable for classification?

→ Logistic Regression is a supervised machine learning algorithm used for classification Problems. It predicts the probability that an observation belongs to a particular class using a logistic (sigmoid) function. It is suitable for classification because it produces outputs between 0 and 1, making it easy to classify data into categories.

② Explain classification performance metrics and why accuracy alone is insufficient.

→ Classification performance metrics include Accuracy, Precision, Recall, F1-Score and AUC-ROC. Accuracy measures overall correctness, but it can be misleading for imbalanced datasets. A model may achieve high accuracy by predicting only the majority class while failing to identify important minority class instances.

③ Define Type-I Error and Type-II Error in the context of risk prediction.

→ Type-I Error (false positive): The model predicts a customer as high risk when the customer is actually low risk.

Type-II Error (False Negative): The model predicts a customer as low risk when the customer is actually high risk. In this Prediction, Type-II Errors are usually more serious because risky customers may go undetected.

4. Explain Precision, Recall, F1-Score, TPR and FPR.

→ Precision : out of all positive predictions, how many were actually positive.

→ Recall (TPR): out of all positive cases, how many were correctly identified.

→ F1-Score : Harmonic mean of precision & Recall.

→ TPR (True positive Rate): Same as Recall.

→ FPR (False Positive Rate): The proportion of negative cases incorrectly clasified as positive.

5. What is AUC-ROC and how does it help in evaluating classifiers?

→ ROC (Receiver operating characteristic) curve plots the True Positive Rate against the False Positive Rate.

AUC (Area Under the curve) measures the overall ability of the model to distinguish between classes.

A higher AUC value indicates better classification performance, with 1.0 representing a perfect classifier.

(6) Why does imbalanced data create problems in classification models?

→ Imbalanced data occurs when one class has significantly more samples than another. classification models tend to favor the majority class, leading to poor detection of the minority class. This can result in misleading accuracy and an increased number of false negatives for important cases.